CYBER THREAT LANDSCAPE BRIEF

Cybersecurity Awareness for College Students

Introduction. College students use the internet for classes, banking, social media, email, and file sharing. These activities are convenient, but can expose personal information, money, and files to cyber threats. Knowing common risks and taking basic precautions can reduce the chance of harm.

1. Phishing Attacks

What it is: Fake emails, texts, or social messages designed to trick people into revealing passwords, banking details, or other sensitive information.

Example/source: CERT-In awareness materials warn about fraudulent messages and links that may lead to fake websites or malicious apps. This is general guidance, not a specific incident.

Warning signs: Urgent messages, unknown senders, suspicious links, spelling mistakes, or requests for passwords/OTPs.

Protect yourself: (1) Check the sender and website address. (2) Never share passwords, PINs, or OTPs in messages. (3) Report suspicious messages and open official sites directly.

2. Weak or Reused Passwords

What it is: Easy-to-guess passwords or reusing one password across accounts can let attackers access multiple accounts if a password is exposed.

Example/source: CERT-In password-security guidance recommends safer password practices to reduce unauthorized access; this is guidance rather than a named breach.

Warning signs: Simple passwords such as 123456, sharing passwords, or using the same password for email and social media.

Protect yourself: (1) Use long, unique passwords. (2) Use a trusted password manager. (3) Enable multi-factor authentication (MFA).

3. Ransomware

What it is: Malicious software that encrypts or locks files and may demand payment to restore access.

Example/source: Verizon’s 2025 Data Breach Investigations Report discusses ransomware as a significant cybersecurity threat; it supports the broader risk, not a specific student incident.

Warning signs: Unexpected file changes, suspicious attachments, unusual device behaviour, or ransom messages.

Protect yourself: (1) Back up files securely. (2) Update operating systems and apps. (3) Avoid unknown attachments, pirated software, and suspicious downloads.

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Practical protections, conclusion, and sources

4. Public Wi-Fi Risks

What it is: Public Wi-Fi may be unsafe. Fraudulent hotspots or poorly secured networks can expose users to phishing pages and other attacks.

Example/source: CERT-In mobile-security guidance advises caution on untrusted networks. This is general guidance, not a documented incident involving a particular student.

Warning signs: Unknown network names, unexpected login pages, or requests for sensitive information to access free Wi-Fi.

Protect yourself: (1) Confirm the network with venue staff. (2) Avoid sensitive transactions on untrusted networks. (3) Disable automatic Wi-Fi connections and use mobile data for sensitive tasks when possible.

5. Fake or Malicious Mobile Applications

What it is: Fake apps imitate trusted apps to steal information, show scams, or install malware. Some request unnecessary permissions.

Example/source: CERT-In advisories on fake mobile applications and mobile malware describe risks from fraudulent or malicious apps; this is general advisory material rather than a named incident.

Warning signs: Apps from unfamiliar sites, excessive permissions, suspicious reviews, or requests to install files outside trusted stores.

Protect yourself: (1) Download from official app stores or verified developers. (2) Check developer details, reviews, and permissions. (3) Update apps and remove apps you no longer trust.

General Cybersecurity Checklist

• Use unique passwords and enable MFA. • Keep devices and apps updated.
• Back up important college documents. • Verify links, QR codes, and attachments.
• Never share OTPs or banking PINs. • Report suspicious activity.

Conclusion

Phishing, weak passwords, ransomware, unsafe public Wi-Fi, and malicious apps can put accounts, personal information, and files at risk. Students can reduce these risks by staying alert, using strong security settings, updating devices, and downloading software from trusted sources. **Think before you click, verify before you trust, and back up before you lose data.**

Sources

1. CERT-In, *Fake Mobile Applications*, CIAD-2018-0033: https://www.cert-in.org.in/s2cMainServlet?VLCODE=CIAD-2018-0033&pageid=PUBVLNOTES02

2. CERT-In, *Securing Mobile Devices and Applications*, CIAD-2020-0013: https://www.cert-in.org.in/s2cMainServlet?VLCODE=CIAD-2020-0013&pageid=PUBVLNOTES02

3. CERT-In, *Mobile-Based Malware: Methods and Countermeasures*, CIAD-2022-0014: https://www.cert-in.org.in/s2cMainServlet?VLCODE=CIAD-2022-0014&pageid=PUBVLNOTES02

4. CERT-In, *Password Management and Security*, CIAD-2022-0026: https://www.cert-in.org.in/s2cMainServlet?pageid=PUBADVLIST

5. CISA, *StopRansomware Guide*: https://www.cisa.gov/stopransomware/ransomware-guide

6. Verizon, *2025 Data Breach Investigations Report*: https://www.verizon.com/business/resources/reports/dbir/

7. CERT-In, *Cyber Security Awareness Booklets and Advisories*: https://www.cert-in.org.in/AwarenessBooklets.jsp

Source note: The examples above cite general guidance or report findings; they should not be interpreted as specific documented incidents unless the cited source identifies one.